Meeting 1015

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國立東華大學應數統計所

October 15, 2025

目錄

資料介紹

SECOM

- 樣本數 1567
- 變數 Time ／ x1 ／ x2 ／... ／ x590 ／ PassFail
  - Features: 591, Y: PassFail (binary classification)
  - Time: 2008-07-19 11:55:00
  - PassFail: Pass $\rightarrow$ 0, Fail $\rightarrow$ 1 (14:1)
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  - 方法
    - Data preprocessing
    - KNN-imputation
    - Random forest features selection
    - imbalance - SMOTE to 1:1
    - 0.8 training, 0.2 testing by decision tree
  - 結果
    - Best: ACC 86.94%, FAR 9.22%
    - 反查 Fail 節點

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研究動機

- K-folds cross validation
  - Split to training&testing first
  - Robust performance
- Metric
  - ACC, FAR, Sensitivity, Specificity
- $y = \hat{f}(x)$
  - Important features

參考資料

Li-Fu Lai (2022)。

使用 KNN 和決策樹方法對填補缺失值的製造數據進行失效預測和原因追蹤分析

The End

Questions & Comments